

Evolutionary Computation and its Applications

Today's goals

Module 1

Fundamental Concepts in Optimization

Module 2

Genetic Algorithm

Module 5 Introduction

Introduction to Module 5's Study

Video • 1 min

How to Access MATLAB Online

Reading • 5 min

Module 5-1 Genetic Algorithm 1

Genetic Algorithm: A First Introduction

Video • 6 min

Genetic Algorithm: A Demo

Video • 8 min

Materials on Genetic Algorithms

Reading • 10 min

How Genetic Algorithms Solve Optimization Problems

Reading • 5 min

Module 5-2 Genetic Algorithm 2

Genetic Algorithms: GA Representation, Initialization and Selection

Video • 10 min

Genetic Algorithms: GA Crossover and Mutation

Video • 7 min

Genetic Algorithms: GA Stopping Criterion and Summary

Video • 5 min

Materials on Genetic Algorithms

Reading • 10 min

How GA Operators Work Together

Reading • 5 min

Module 5-3 Use Genetic Algorithm 1

Use GA via MATLAB: GA Function

Video • 7 min

Use GA via MATLAB: Employ the Function

Video • 6 min

Materials on Using Genetic Algorithms

Reading • 10 min

Using Genetic Algorithms Effectively in MATLAB

Reading • 5 min

Module 5-4 Use Genetic Algorithm 2

Use GA via MATLAB: Case Studies

Video • 5 min

Materials on Case Studies

Reading • 10 min

Solving Real Engineering Problems with Genetic Algorithms

Reading • 5 min

Module 5 Quiz

Module 5 Quiz

Practice Assignment • Grade: 100%

Module 5 Assignment

Module 5 Assignment

Graded App Item • 30 min

Objective function for Assignment 5

Reading • 5 min

Applying Genetic Algorithms to a Real Engineering Design Problem

Reading • 5 min

Designing and Applying a Genetic Algorithm to Solve an Engineering Optimization Problem

Dialogue • 5 min

Module 5 Recap

Reading • 5 min

Module 3

Particle Swarm Optimization

Module 5 Recap

0:01 / 1:49

2x

In this module, you learned the foundational principles of Genetic Algorithms (GA), including how evolutionary concepts such as population-based search, selection, crossover, mutation, and fitness evaluation are used to solve complex optimization problems. You also explored how GA operators work together, how GA is implemented in MATLAB, and how genetic algorithms can be applied to real engineering and scientific optimization problems.

Reflect on the following questions:

- What were the most important ideas you learned about how genetic algorithms search for and improve candidate solutions over generations?
- Why is maintaining a balance between exploration and exploitation important in genetic algorithms, and how do operators such as selection, crossover, and mutation contribute to this balance?
- Based on what you learned about GA representation, initialization, selection pressure, crossover, mutation, and stopping criteria, what factors do you think most strongly influence whether a genetic algorithm converges successfully or becomes trapped in poor solutions?
- How does genetic algorithm optimization differ from conventional or local optimization methods, particularly when solving multimodal, high-dimensional, or poorly understood engineering problems?
- After working with MATLAB implementations of GA, why is it important to understand optimization behavior, convergence, parameter settings, and constraints rather than treating optimization software as a black box?

This discussion will help you consolidate your understanding of how genetic algorithms operate internally, how they are implemented in practice, and why evolutionary optimization methods are powerful tools for solving complex real-world engineering and scientific problems.

Go to next item

✓ Completed

Like

Dislike

Report an issue